

Titration report

LabX

Method

Method ID	CDC_GranTED-DYN-KHP-M002	Release state	Released
Name	Titration KHP with NaOH based on M002	Modified by	PredefinedUser
Modified on	3/9/2026 7:46:45 AM		

Summary

Samples

	Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.034797 mL	OK
		Concentration R2	0.10070 mol/L	OK
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.029781 mL	OK
		Concentration R2	0.10060 mol/L	OK
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.034484 mL	OK
		Concentration R2	0.10069 mol/L	OK

Statistics

R1

Mean	5.033021 mL	Minimum	5.029781 mL
Standard deviation	0.002810 mL	Rel. standard deviation	0.056 %
Total samples	3	Samples excluded	0

R2

Mean	0.10066 mol/L	Minimum	0.10060 mol/L
Standard deviation	0.00006 mol/L	Rel. standard deviation	0.055 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165503	Task name	DYN 0.02-0.2 mL VDISP=2mL - M002 - KHP/NaOH
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.034797 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10070 mol/L	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.029781 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10060 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.034484 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10069 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/9/2026 7:47:56 AM
Overall state	OK
Sample start time	3/9/2026 7:47:56 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/9/2026 7:47:57 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/9/2026 7:47:57 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.50 mmol
tDi	00:08 min

Stir (Stir1)

Creation time	3/9/2026 7:48:14 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	3/9/2026 7:48:30 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	-103.6 mV
EHNv1	94.7 mV
Q1	0.503480 mmol
VEQ1	5.034797 mL
EST	159.1 mV
VEX	0.649203 mL
VEND	5.6840 mL
t	02:50 min
QEND	0.568400 mmol
QEX	0.064920 mmol

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EPD 104.3 mV
nEQ 1

Consumption (R1)

Creation time 3/9/2026 7:51:25 AM
Overall state OK
Result 5.034797 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

Creation time 3/9/2026 7:51:25 AM
Overall state OK
Result 0.10070 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time 3/9/2026 7:51:55 AM
Overall state OK
Sample start time 3/9/2026 7:51:55 AM
Sample size 5 mL
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 3/9/2026 7:51:56 AM
Overall state OK

Dispense (normal) [1] (Dispensenormal1)

Creation time 3/9/2026 7:51:56 AM
Overall state OK
VENDDi 5 mL
QENDDi 0.50 mmol
tDi 00:08 min

Stir (Stir1)

Creation time 3/9/2026 7:52:14 AM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 3/9/2026 7:52:29 AM
Overall state OK
Terminated at Number of EQPs

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EEQ1	-102.7 mV
EHN1	95.1 mV
Q1	0.502978 mmol
VEQ1	5.029781 mL
EST	159.4 mV
VEX	0.422719 mL
VEND	5.4525 mL
t	02:52 min
QEND	0.545250 mmol
QEX	0.042272 mmol
EPD	104.6 mV
nEQ	1

Consumption (R1)

Creation time	3/9/2026 7:55:25 AM
Overall state	OK
Result	5.029781 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	3/9/2026 7:55:26 AM
Overall state	OK
Result	0.10060 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/9/2026 7:55:48 AM
Overall state	OK
Sample start time	3/9/2026 7:55:48 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/9/2026 7:55:49 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/9/2026 7:55:49 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.50 mmol
tDi	00:08 min

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Stir (Stir1)

Creation time 3/9/2026 7:56:06 AM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 3/9/2026 7:56:21 AM
Overall state OK
Terminated at Number of EQPs
EEQ1 -97.0 mV
EHN1 94.4 mV
Q1 0.503448 mmol
VEQ1 5.034484 mL
EST 158.9 mV
VEX 0.455516 mL
VEND 5.4900 mL
t 02:53 min
QEND 0.549000 mmol
QEX 0.045552 mmol
EPD 104.2 mV
nEQ 1

Consumption (R1)

Creation time 3/9/2026 7:59:20 AM
Overall state OK
Result 5.034484 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

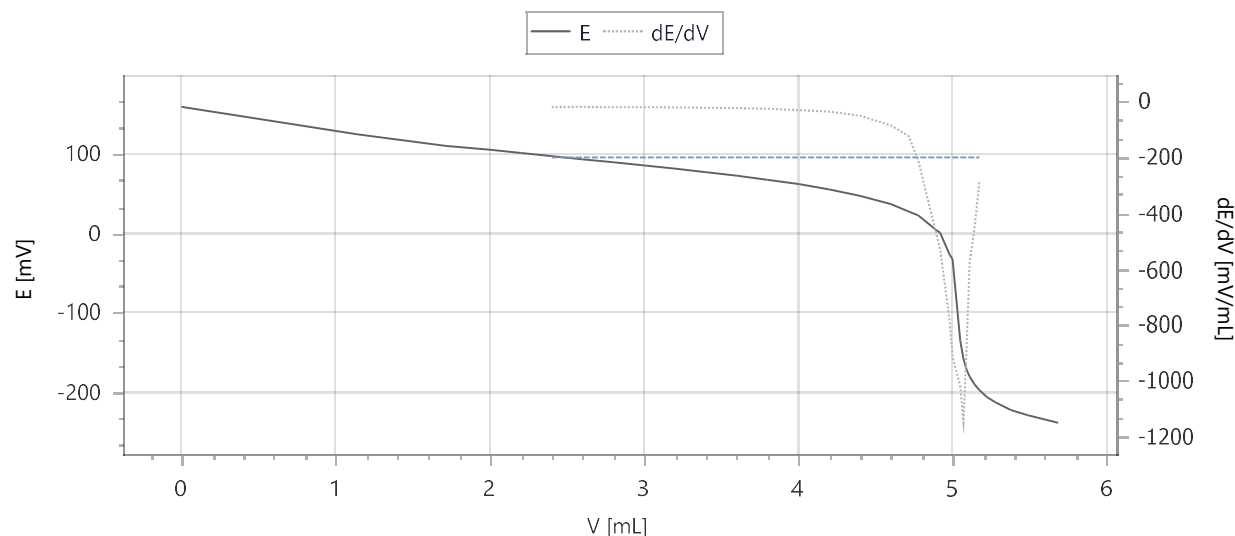
Creation time 3/9/2026 7:59:20 AM
Overall state OK
Result 0.10069 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	159.1		25.0			
5	1.1430	124.4	1.1430	25.0	-34.7		
9	1.7145	110.2	0.5715	25.0	-14.2		
12	2.0000	105.3	0.2855	25.0	-4.9		
46	2.2000	101.0	0.2000	25.0	-4.3		
48	2.4000	97.0	0.2000	25.0	-4.0	-19.40	
52	2.6000	93.1	0.2000	25.0	-3.9	-18.96	
55	2.8000	89.5	0.2000	25.0	-3.6	-19.64	
58	3.0000	85.4	0.2000	25.0	-4.1	-19.77	
61	3.2000	81.6	0.2000	25.0	-3.8	-20.60	
64	3.4000	77.0	0.2000	25.0	-4.6	-21.86	
67	3.6000	72.5	0.2000	25.0	-4.5	-23.43	
70	3.8000	67.5	0.2000	25.0	-5.0	-25.57	
73	4.0000	62.1	0.2000	25.0	-5.4	-30.24	
76	4.2000	55.3	0.2000	25.0	-6.8	-35.73	
79	4.4000	47.4	0.2000	25.0	-7.9	-50.68	
82	4.6000	36.8	0.2000	25.0	-10.6	-85.35	
85	4.7150	27.3	0.1150	25.0	-9.5	-124.06	
88	4.7765	22.7	0.0615	25.0	-4.6	-209.37	
94	4.8945	4.3	0.1180	25.0	-18.4	-474.50	
97	4.9190	1.0	0.0245	25.0	-3.3	-530.49	
100	4.9800	-27.1	0.0610	25.0	-28.1	-784.38	
103	5.0000	-32.5	0.0200	25.0	-5.4	-917.64	
	5.034797	-103.6				-1209.26	EQP
118	5.0500	-134.7	0.0500	25.0	-102.2	-1022.10	
124	5.0700	-157.9	0.0200	25.0	-23.2	-1151.95	
129	5.0900	-171.4	0.0200	25.0	-13.5	-885.83	
133	5.1100	-179.9	0.0200	25.0	-8.5	-580.70	
136	5.1405	-189.5	0.0305	25.0	-9.6	-459.78	
139	5.1745	-197.3	0.0340	25.0	-7.8	-283.90	
142	5.2230	-206.1	0.0485	25.0	-8.8		

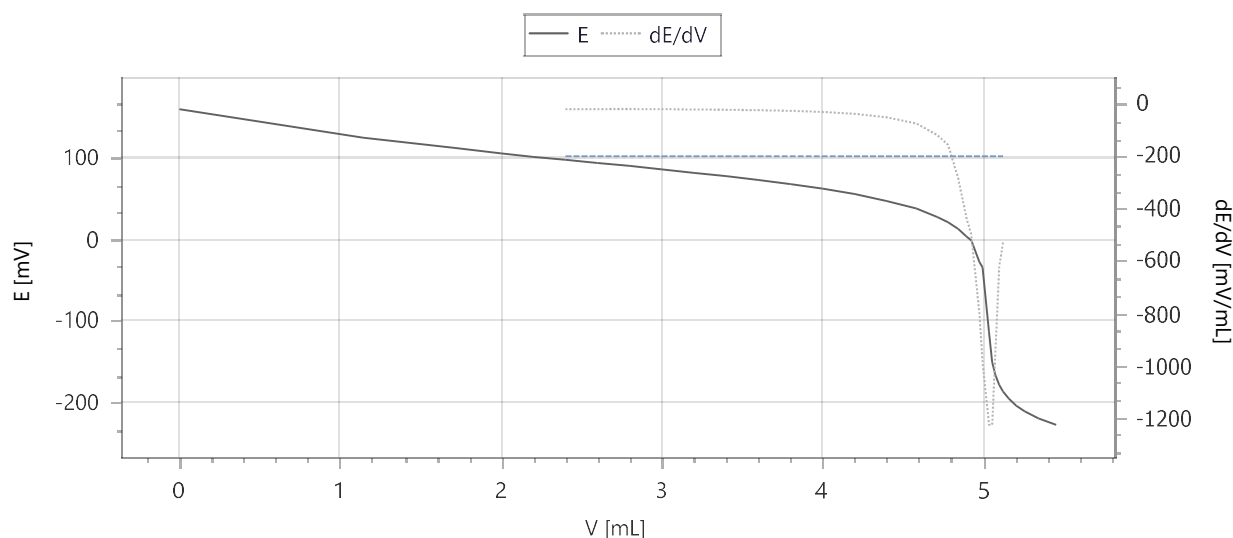
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146	5.2780	-212.8	0.0550	25.0	-6.7
148	5.3775	-222.3	0.0995	25.0	-9.5
152	5.4845	-228.9	0.1070	25.0	-6.6
155	5.6840	-238.4	0.1995	25.0	-9.5

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	159.4		25.0			
6	1.1430	124.2	1.1430	25.0	-35.2		
9	1.7145	111.6	0.5715	25.0	-12.6		
12	2.0000	105.3	0.2855	25.0	-6.3		
45	2.2000	101.0	0.2000	25.0	-4.3		
48	2.4000	97.3	0.2000	25.0	-3.7	-19.53	
52	2.6000	93.4	0.2000	25.0	-3.9	-19.56	
55	2.8000	89.7	0.2000	25.0	-3.7	-19.35	
58	3.0000	85.7	0.2000	25.0	-4.0	-20.03	
61	3.2000	81.4	0.2000	25.0	-4.3	-20.99	
64	3.4000	77.2	0.2000	25.0	-4.2	-22.01	
67	3.6000	72.5	0.2000	25.0	-4.7	-23.83	
70	3.8000	67.5	0.2000	25.0	-5.0	-25.99	
73	4.0000	61.9	0.2000	25.0	-5.6	-30.31	
76	4.2000	55.2	0.2000	25.0	-6.7	-37.68	
79	4.4000	46.8	0.2000	25.0	-8.4	-51.12	
82	4.5835	37.6	0.1835	25.0	-9.2	-74.62	
86	4.7145	27.3	0.1310	25.0	-10.3	-119.26	
89	4.7795	20.8	0.0650	25.0	-6.5	-156.18	
92	4.8435	12.8	0.0640	25.0	-8.0	-284.20	
96	4.8940	3.6	0.0505	25.0	-9.2	-433.41	
99	4.9245	-1.5	0.0305	25.0	-5.1	-498.90	
102	4.9755	-27.9	0.0510	25.0	-26.4	-802.42	

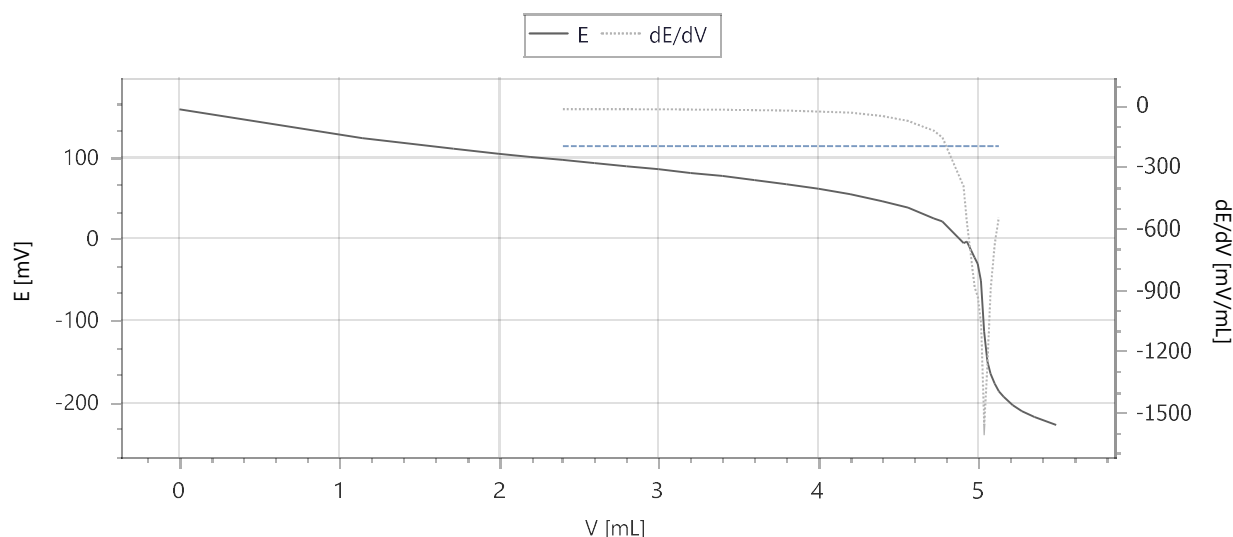
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105	4.9955	-34.4	0.0200	25.0	-6.5	-991.65	EQP
	5.029781	-102.7				-1262.70	
121	5.0350	-113.1	0.0395	25.0	-78.7	-1225.56	
130	5.0550	-150.4	0.0200	25.0	-37.3	-1227.02	
135	5.0750	-166.3	0.0200	25.0	-15.9	-943.54	
139	5.0985	-178.3	0.0235	25.0	-12.0	-621.91	
142	5.1230	-186.6	0.0245	25.0	-8.3	-525.60	
145	5.1580	-195.1	0.0350	25.0	-8.5		
148	5.2040	-203.9	0.0460	25.0	-8.8		
151	5.2575	-211.0	0.0535	25.0	-7.1		
154	5.3430	-219.5	0.0855	25.0	-8.5		
157	5.4525	-227.4	0.1095	25.0	-7.9		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	158.9		25.0			
5	1.1430	123.4	1.1430	25.0	-35.5		
9	1.7145	110.4	0.5715	25.0	-13.0		
12	2.0000	104.3	0.2855	25.0	-6.1		
45	2.2000	100.3	0.2000	25.0	-4.0		
48	2.4000	96.6	0.2000	25.0	-3.7	-19.26	
52	2.6000	92.8	0.2000	25.0	-3.8	-19.18	
55	2.8000	88.8	0.2000	25.0	-4.0	-19.09	
58	3.0000	85.1	0.2000	25.0	-3.7	-19.80	
61	3.2000	80.6	0.2000	25.0	-4.5	-20.80	
64	3.4000	77.0	0.2000	25.0	-3.6	-21.72	
67	3.6000	72.1	0.2000	25.0	-4.9	-23.86	
70	3.8000	66.7	0.2000	25.0	-5.4	-25.80	
73	4.0000	61.3	0.2000	25.0	-5.4	-30.80	
76	4.2000	54.8	0.2000	25.0	-6.5	-35.72	

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79	4.4000	45.9	0.2000	25.0	-8.9	-52.36	EQP
82	4.5620	38.3	0.1620	25.0	-7.6	-76.76	
85	4.7245	24.7	0.1625	25.0	-13.6	-124.54	
88	4.7775	21.1	0.0530	25.0	-3.6	-157.20	
91	4.9100	-5.5	0.1325	25.0	-26.6	-395.67	
94	4.9300	-3.3	0.0200	25.0	2.2	-563.34	
98	4.9800	-23.1	0.0500	25.0	-19.8	-891.79	
102	5.0000	-31.1	0.0200	25.0	-8.0	-939.81	
109	5.0200	-51.8	0.0200	25.0	-20.7	-1069.10	
	5.034484	-97.0				-1566.07	
123	5.0400	-114.2	0.0200	25.0	-62.4	-1564.33	
132	5.0600	-150.5	0.0200	25.0	-36.3	-1262.06	
136	5.0800	-165.8	0.0200	25.0	-15.3	-898.75	
140	5.1050	-178.0	0.0250	25.0	-12.2	-683.44	
143	5.1305	-187.2	0.0255	25.0	-9.2	-548.46	
146	5.1610	-194.0	0.0305	25.0	-6.8		
149	5.2175	-204.0	0.0565	25.0	-10.0		
152	5.2755	-211.8	0.0580	25.0	-7.8		
156	5.3530	-218.7	0.0775	25.0	-6.9		
159	5.4900	-228.6	0.1370	25.0	-9.9		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=2mL - M002 - KHP/NaOH	09.03.2026 07:47:54

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0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=2mL - M002 - KHP/NaOH	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=2mL - M002 - KHP/NaOH	09.03.2026 07:47:54
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:47:54

Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
09.03.2026 07:47:54	PredefinedUser	Administrator